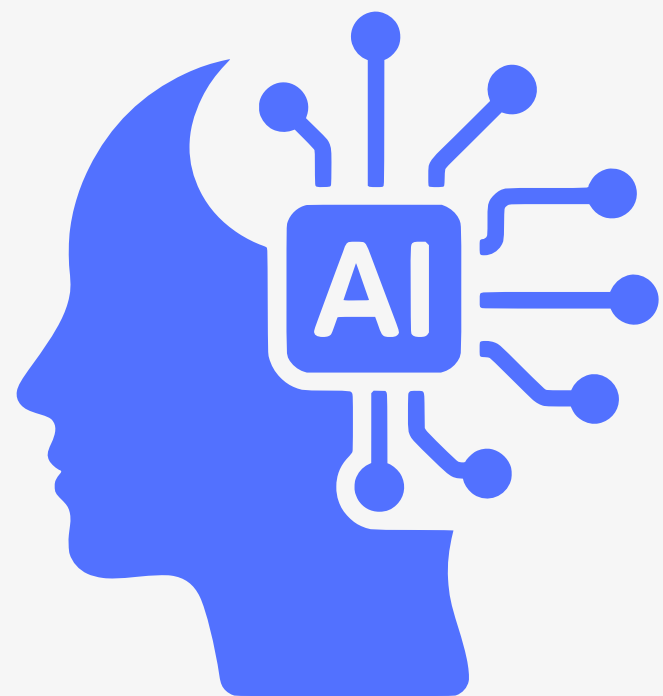




EVAL-DRIVEN RAG SYSTEMS

LLMS DON'T KNOW YOUR COMPANY'S DATA. RAG IS WHAT GIVES
THEM KNOWLEDGE.





WHAT IS RAG?

- RAG = Retrieval-Augmented Generation
- Combines information retrieval with LLMs
- Retrieves relevant knowledge before generating an answer
- Reduces hallucinations
- Enables AI to answer from private or custom data
- Foundation of modern enterprise AI applications





WHY DO WE NEED RAG?

- LLMs cannot know your private company data
- Training an LLM for every document is expensive
- Company data changes frequently
- RAG keeps knowledge up to date
- No model retraining required





WHERE IS RAG USED?

- Enterprise AI Assistants
- Customer Support Chatbots
- Internal Company Knowledge Bases
- Legal AI
- Medical AI
- Banking AI
- HR Assistants
- PDF Chat Applications





RAG ARCHITECTURE OVERVIEW

- User asks a question
- Retrieve relevant information
- Build context
- Send context to LLM
- Generate accurate response





RETRIEVAL PIPELINE

- Receive user query
- Convert query into embeddings
- Search similar chunks
- Retrieve top relevant documents
- Send results to the LLM





WHAT ARE EMBEDDINGS?

- Numerical representation of text
- Captures semantic meaning
- Similar meanings have similar vectors
- Used for semantic search
- Foundation of vector databases





CONTEXT ASSEMBLY

- Combine retrieved chunks
- Build final prompt
- Preserve important context
- Send only relevant information
- Improve answer quality





RAG VS FINE-TUNING

RAG

-
- Uses external knowledge
- Easy to update
- Lower cost
- No retraining

Fine-Tuning

-
- Changes model behavior
- Expensive
- Requires training
- Best for specialized tasks





BENEFITS OF RAG

- More accurate responses
- Fewer hallucinations
- Private knowledge support
- Lower AI costs
- Easy knowledge updates
- Production-ready architecture

